Timing Search[†]

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Abstract

Why do different industries fluctuate at different frequencies? I introduce *timing search*—agents choosing when to enter a market under frictions that penalize rapid adjustment. In equilibrium, agents are indifferent across entry dates, yielding an *ironing condition along time* that produces a conservation law with cycle frequency $\omega = \sqrt{\varphi/\mu}$, where φ is the curvature of the profit landscape and μ is the timing friction, micro-founded from congestion in the entry matching process. The frequency depends only on market structure, not on the shock process. I confirm this cross-sectionally—geographic concentration predicts cycle frequency across 16 industries—and dynamically: after large historical shocks, industries display statistically significant deterministic oscillations at their structural frequency, which gradually dissipate.

Keywords: Timing search, cycle frequency, market structure, ironing, scale externality.

JEL codes: D83, E32, L13, R12.

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1 Introduction

Industries fluctuate at different frequencies. Technology clusters cycle in rapid three-to-five-year waves, while commodity industries roll through slow seven-to-ten-year swings (Dunne et al., 1988). Yet no existing framework explains why. Standard models derive cycle properties from a common shock process (Kydland and Prescott, 1982) or policy rule (Christiano et al., 2005), leaving cross-industry variation in frequency unexplained. This paper introduces *timing search*—agents choosing *when* to enter a market under frictions that prevent instantaneous adjustment—and shows that the equilibrium cycle frequency is determined by two market-structure primitives:

$$\omega = \sqrt{\frac{\varphi}{\mu}}, \tag{1}$$

where φ is the *curvature of the profit landscape*—the sharpness of the scale–congestion trade-off—and μ is the *timing friction*—the cost of rapid aggregate adjustment. Both are industry-specific and independently measurable.

The mechanism is as follows. Profitability is hump-shaped in the number of active firms, peaking at an intermediate density n_p. Agents who cannot adjust instantaneously face a timing problem: choosing when to enter. In equilibrium, no agent can gain by shifting her entry to a different date. Otherwise, agents would crowd the profitable dates, eliminating the advantage. This *ironing condition along time*—the requirement that effective payoffs be equalized across all entry dates—is the temporal analog of the equal-profit condition in Burdett and Mortensen (1998). Combined with the multiplicative timing friction, it produces a conservation law whose periodic orbits constitute the equilibrium business cycle, with exact solution $n(t) = n_p + A\cos(\omega t + \phi)$.

The paper makes four contributions. First, I introduce timing search as a new equilibrium concept for industry dynamics (Sections 2–3). The concept extends search theory from the cross-section (wages, prices) to the time domain (entry dates). The ironing condition along time is the central equilibrium object. I show that it generates endogenous business cycles through a conservation law: the equilibrium orbit is an ellipse in the $(n, \dot{n})$ phase plane, traversed at frequency $\omega = \sqrt{\varphi/\mu}$, with no linearization required. The timing friction μ is micro-founded from congestion in the entry matching process (Section 2.3), connecting to the DMP literature (Pissarides, 2000; Petrongolo and Pissarides, 2001).

Second, I provide a structural decomposition of cycle frequency. Equation (1) decomposes ω into two independently measurable primitives: profit-landscape curvature φ and the timing friction μ. Cycle frequency is therefore independent of the shock process, productivity levels, and cycle amplitude—a property I call *isochrony*. No existing model delivers such a decomposition. The Gaussian profit specification is quantitatively accurate: the frequency correction under general profits is $O(A^2)$ and below 5% for calibrated amplitudes (Proposition 4). Non-Gaussian profits generate asymmetric cycles, with the degree of asymmetry pinned by measurable profit-landscape skewness (Proposition 5).

Third, in the stochastic extension (Section 4), small i.i.d. shocks produce a stochastic limit cycle whose spectral peak is pinned at $\omega = \sqrt{\varphi/\mu}$. The shock process governs amplitude but not frequency. Adding entry-cost dissipation (Section 4.4) resolves the free-amplitude indeterminacy: the balance of forcing and dissipation selects a unique amplitude $A_{\text{rms}} = \sigma_\varepsilon/\sqrt{2\eta\varphi\mu}$, explaining both cycle frequency and severity from primitives.

Fourth, I take the cross-industry predictions to U.S. data (Section 5). Monthly industrial production indices for eight sectors over 1972–2024 reveal substantial variation in spectral peak periods (2.6–6.6 years). Independently measured market-structure proxies explain 75% of this variation, with both coefficient signs matching the model. I also derive a welfare cost of endogenous cycles: $\mathcal{L} \approx \frac{\varphi}{2}A^2$, connecting to Lucas (1987) with a structural answer (Section 6.6).

Related literature This paper contributes to five strands of literature.

The first is endogenous business cycles. Beaudry et al. (2020) generate a stochastic limit cycle from financial frictions via Hopf bifurcation. Beaudry and Portier (2006) showed that stock prices lead TFP, consistent with endogenous boom–bust dynamics. Grandmont (1985) established the possibility of endogenous competitive business cycles. The present mechanism is orthogonal: scale–congestion rather than financial frictions. The key difference is that ω here is a closed-form function of two primitives, whereas in Beaudry et al. (2020) it is determined implicitly by a bifurcation condition. The two mechanisms could coexist.

The second is search theory. Timing search extends Stigler (1961), McCall (1970), and Burdett and Mortensen (1998) to the time domain. In standard search, agents choose among contemporaneous alternatives (wages, prices, jobs); here, they choose

among dates. The deepest connection is to Burdett and Mortensen (1998): both models rest on an equal-profit condition—wages in their case, entry dates in mine. Online Appendix A develops the formal one-sided problem.

Third, on industry dynamics, Hopenhayn (1992) models entry and exit with full information; I add timing frictions. Jovanovic (1982) showed that learning about fixed types generates the industry cross-section without aggregate shocks. Timing search generates the aggregate *dynamics* without aggregate shocks. The two are complementary: Jovanovic (1982) explains endogenous exit through type discovery; timing search explains endogenous entry waves through the ironing condition.

Fourth, Krugman (1991) showed that non-monotone externalities generate spatial concentration. Duranton and Puga (2004) and Rosenthal and Strange (2004) provide empirical evidence for scale and congestion forces. Ellison and Glaeser (1997a) and Ellison et al. (2010) measure geographic concentration across manufacturing, providing empirical counterparts for φ. I transpose the scale–congestion logic from space to time: combined with timing frictions, it generates temporal patterns (cycles) rather than spatial patterns (clusters).

Fifth, the micro-foundation of μ from matching congestion connects to the Diamond-Mortensen-Pissarides literature (Pissarides, 2000; Petrongolo and Pissarides, 2001). In DMP models, matching frictions generate equilibrium unemployment. Here, matching frictions in entry generate equilibrium business cycles. The formula $\mu = (1 - \alpha)/[\alpha(\delta n_p)^2]$ maps the matching elasticity α to cycle frequency.

2 Environment

2.1 Agents

Assumption 1 (Entry economy)**.** *A mass $M > 0$ of agents, each at any instant either* active *(operating a firm) or* searching *(a potential entrant). Active firms produce using labor with technology $y_i = zl_i^\gamma$, $\gamma \in (0, 1)$, and exit at Poisson rate $\delta > 0$, returning to the search pool. A representative household supplies labor according to $w = \eta L_h$ with $\eta > 0$ (GHH preferences, Frisch elasticity of 1).*

Remark 1 (Role of the preference and exit assumptions)**.**
GHH preferences eliminate wealth effects on labor supply, so the profit landscape depends only on n, not on the distribution of wealth. Exogenous exit at rate δ is standard

(Hopenhayn, 1992). It ensures a steady flow of agents returning to the search pool, maintaining the timing search problem at every date. Section 6 discusses relaxations.

Let $n(t)$ denote the mass of active firms at date t and $S(t) = M - n(t)$ the mass of searchers.

2.2 Matching efficiency and the profit landscape

Each firm's effective output depends on market thickness. More firms create denser networks of suppliers, customers, and complementary services, improving efficiency. But beyond a point, additional firms congest the market. A scale efficiency function $m(n)$ multiplies each firm's output: $y_i = zl_i^\gamma \cdot m(n)$. Profit is $\pi(n) = \bar{\pi}(w) \cdot m(n)$, where $\bar{\pi}$ is the standard profit given the wage.

Assumption 2 (Gaussian scale efficiency)**.** *Scale efficiency takes the Gaussian form:*

$$m(n) = \exp\Big(-\frac{\varphi}{2}(n - n_p)^2\Big), \qquad \varphi > 0,\ n_p > 0. \tag{2}$$

Remark 2 (Interpretation of Assumption 2)**.**
The function peaks at $n = n_p$ (optimal market scale) and falls off symmetrically. The parameter φ governs the sharpness of the trade-off: high φ means a narrow peak (sharp trade-off); low φ means a broad peak (robust to deviations). Online Appendix B provides a matching-theoretic microfoundation.

For the analytical results, I work with the reduced-form Gaussian profit:

$$\pi(n) = \pi_0 \exp\Big(-\frac{\varphi}{2}(n - n_p)^2\Big), \qquad \pi_0 > 0. \tag{3}$$

The semi-elasticity is exactly linear:

$$f(n) \equiv \frac{\pi'(n)}{\pi(n)} = -\varphi(n - n_p). \tag{4}$$

This linearity delivers exact harmonic oscillations (Theorem 1). The general case—where wage congestion adds a non-quadratic correction to $\ln \pi$—produces anharmonic oscillations with the same leading-order frequency (Proposition 3).

2.3 The timing friction: congestion in the entry process

Entry is mediated by a matching function. At each instant, the $S(t) = M - n(t)$ searchers compete for R units of *entry resources*—permits, specialized labor, venture capital, factory sites—to form matches. Successful matches become active firms.

Assumption 3 (Entry matching)**.** *The flow of new matches is*

$$m(S, R) = S^{\alpha} R^{1-\alpha}, \qquad \alpha \in (0, 1). \tag{5}$$

Each searcher's matching probability per unit time is $p(\theta_e) = \theta_e^{-(1-\alpha)}$, where $\theta_e \equiv S/R$ is entry-market tightness. When many firms try to enter simultaneously ($\dot{n}$ large), the entry flow $e = \delta n + \dot{n}$ spikes, tightness rises, and the matching probability per searcher falls. This is the timing friction: congestion in the entry process.

Proposition 1 (Micro-foundation of the timing friction)**.** *Under Assumptions 1–3, the entry matching friction produces, to second order around the steady state, the timing friction* $\xi(\dot{n}) = e^{-\frac{\mu}{2}\dot{n}^2}$ *with*

$$\mu = \frac{1-\alpha}{\alpha\,(\delta n_p)^2}. \tag{6}$$

Proof. At steady state, the entry flow is $e_{ss} = \delta n_p$ and the matching probability is $p_{ss} = p(\theta_e^{ss})$. When $\dot{n}$ deviates from zero, the entry flow becomes $e = \delta n_p + \dot{n}$ (to leading order in perturbations). The matching probability for a given searcher depends on the flow tightness: $p(\theta_e) \propto e^{-(1-\alpha)/\alpha}$. The effective friction is the ratio of actual to steady-state matching efficiency:

$$\ln \xi(\dot{n}) = \ln p(\theta_e(\dot{n})) - \ln p(\theta_e^{ss}) = -\frac{1-\alpha}{\alpha} \ln\left(1 + \frac{\dot{n}}{\delta n_p}\right).$$

Expanding to second order: $\ln(1+z) = z - z^2/2 + O(z^3)$. The first-order term (linear in $\dot{n}$) shifts the equilibrium level and is absorbed into the constant C of the ironing condition. The second-order term gives:

$$\ln \xi(\dot{n}) \approx -\frac{1-\alpha}{2\alpha(\delta n_p)^2}\,\dot{n}^2,$$

which is $-\frac{\mu}{2}\dot{n}^2$ with $\mu = (1-\alpha)/[\alpha(\delta n_p)^2]$. ■

Remark 3 (Structural interpretation).
The timing friction μ is: (i) decreasing in α—industries where searchers have higher matching power face less congestion; (ii) decreasing in δn_p—industries with larger steady-state turnover absorb entry waves more easily; (iii) increasing in $1/R$ implicitly through the matching function—industries with scarcer entry resources have higher μ. The matching elasticity α has been estimated in the labor matching literature: Petrongolo and Pissarides (2001) survey estimates and find $\alpha \in [0.5, 0.7]$ for labor markets. For entry matching, α may differ, but (6) provides a mapping from any estimate to μ.

Remark 4 (Structural formula for ω).
Substituting (6) into $\omega = \sqrt{\varphi/\mu}$ yields:

$$\omega = \delta n_p \sqrt{\frac{\alpha\varphi}{1-\alpha}}, \tag{7}$$

connecting the cycle frequency to three independently measurable objects: the matching elasticity α, the profit curvature φ, and the steady-state turnover δn_p.

Effective output at date t is $y(t) = \pi(n(t)) \cdot \xi(\dot{n}(t))$, where $\xi(\dot{n}) = e^{-\frac{\mu}{2}\dot{n}^2}$ and μ is given by (6). The friction is multiplicative: matching congestion scales with the level of profits.

Remark 5 (Robustness of the quadratic form).
The quadratic form in $\dot{n}$ arises generically from the second-order expansion of any smooth, symmetric matching friction around the steady state. Online Appendix C provides an alternative derivation from discrete-time proportional adjustment costs, yielding the same $\xi = e^{-\frac{\mu}{2}\dot{n}^2}$. The results depend on the quadratic structure, not on the specific micro-origin (Cooper and Haltiwanger, 2006).

2.4 Combined structure

Combining the Gaussian profit and the Gaussian friction, effective output is:

$$y(t) = \pi_0 \exp\Big(-\frac{\varphi}{2}(n - n_p)^2 - \frac{\mu}{2}\dot{n}^2\Big). \tag{8}$$

The effective output is a bivariate Gaussian kernel in $(n, \dot{n})$, centered at $(n_p, 0)$. Both terms impose quadratic penalization of deviation from the optimum: n_p for market

scale and $\dot{n} = 0$ for adjustment speed.

3 Ironing Along Time: Equilibrium and the Main Result

3.1 The agent's problem

A searching agent chooses a date τ at which to enter. Upon entry, she operates a firm earning effective flow payoff $\pi(n(t))\xi(\dot{n}(t))$ per unit time, discounted at rate $r \geq 0$, with exogenous exit at rate $\delta > 0$. The value of entering at date τ is:

$$V(\tau) = \int_{\tau}^{\infty} e^{-(r+\delta)(t-\tau)}\, \pi(n(t))\, \xi(\dot{n}(t))\, \mathrm{d}t. \tag{9}$$

The agent maximizes $V(\tau)$ over τ, taking the aggregate path $\{n(t)\}$ as given. An agent who does not enter at date t remains in the search pool, with continuation value

$$V^{\text{search}}(t) = \max_{\tau \geq t} e^{-r(\tau-t)} \big[V(\tau) - \kappa\big], \tag{10}$$

where $\kappa \geq 0$ is the entry cost.

3.2 Equilibrium

An *entry policy* is a measurable function $e : \mathbb{R}_+ \to \mathbb{R}_+$ specifying the flow rate of entrants at each date.

Definition 1 (Timing search equilibrium)**.** *A* timing search equilibrium *is a path* $\{n(t)\}_{t\geq 0}$ *and an entry policy* $e(\cdot)$ *such that:*

(i) ***Individual optimality****: Each agent chooses her entry date to maximize* $V(\tau)$*.*

(ii) ***Temporal free entry****: At every date where entry occurs,* $V(t) = V^{search}(t) + \kappa$*.*

(iii) ***No profitable timing deviation****:* $V(\tau) \leq V(t)$ *for all* τ *and all* t *in the support of entry.*

(iv) ***Aggregate consistency****:* $\dot{n}(t) = e(t) - \delta\, n(t)$*, where* $e(t) \geq 0$*.*

(v) ***Feasibility****:* $n(t) \leq M$ *for all* t*.*

Remark 6 (The no-profitable-timing-deviation condition)**.**
Condition (iii) is the key requirement. In Burdett and Mortensen (1998), the analogous condition is that no firm can profit by reposting its wage, which yields equal profits across wages. Here, the condition that no agent can profit by retiming her entry yields equal profits across dates. If any date offered strictly higher value, agents would crowd that date, changing $n(t)$ until the advantage was eliminated.

3.3 The ironing condition

Proposition 2 (Ironing condition)**.** *In a timing search equilibrium where entry occurs at all dates, the effective flow payoff is equalized:*

$$\pi(n(t))\,\xi(\dot{n}(t)) = C \qquad \textit{for all } t, \tag{11}$$

for some constant $C > 0$.

Proof. Suppose entry occurs at all dates. For any two dates τ_1, τ_2 in the support of entry, condition (iii) of Definition 1 requires $V(\tau_1) \leq V(\tau_2)$ and $V(\tau_2) \leq V(\tau_1)$, so $V(\tau_1) = V(\tau_2)$. Since the support is all of $\mathbb{R}_+$, $V(\tau) = V^*$ for all $\tau \geq 0$. The equilibrium path $n(\cdot)$ is C^1 by aggregate consistency (condition (iv)) and the smoothness of π and ξ, so $V(\tau)$ is differentiable in τ.[1] Differentiating (9):

$$\frac{\mathrm{d}V}{\mathrm{d}\tau} = -\pi(n(\tau))\,\xi(\dot{n}(\tau)) + (r+\delta)\,V(\tau).$$

Setting $\frac{\mathrm{d}V}{\mathrm{d}\tau} = 0$ (since V is constant): $\pi(n(\tau))\xi(\dot{n}(\tau)) = (r+\delta)V^* \equiv C$. ■

Remark 7 (Regularity condition)**.**
The condition that entry occurs at all dates is satisfied whenever the searcher pool $S(t) = M - n(t)$ is bounded away from zero along the equilibrium path and the timing friction μ is positive. With a large population M and moderate cycle amplitude $A < M - n_p$, the condition holds. Online Appendix F provides a formal verification.

[1] The differentiability of V in τ requires that $t \mapsto \pi(n(t))\xi(\dot{n}(t))$ is locally integrable along the equilibrium path, which holds whenever $n(\cdot) \in C^1$. This regularity is guaranteed by the smoothness of π (Assumption 2) and ξ (Section 2.3), combined with the continuity of the entry flow $e(\cdot)$ in condition (iv).

3.4 The conservation law

Taking logarithms of (11) and using (8):

$$-\frac{\varphi}{2}(n-n_p)^2 - \frac{\mu}{2}\,\dot{n}^2 = E, \tag{12}$$

where $E = \ln C - \ln \pi_0 < 0$ is constant along the equilibrium path. This is a conservation law: the sum of the "potential energy" $-\frac{\varphi}{2}(n-n_p)^2$ and "kinetic energy" $-\frac{\mu}{2}\dot{n}^2$ is constant. Geometrically, it is the equation of an ellipse in the $(n, \dot{n})$ phase plane.

3.5 The main result

Differentiating (12) with respect to t and dividing by $\dot{n} \neq 0$:

$$\ddot{n} = -\frac{\varphi}{\mu}\,(n-n_p). \tag{13}$$

Theorem 1 (Exact harmonic oscillations)**.** *In a timing search equilibrium (Definition 1) under Assumptions 1 and 2, the equilibrium path is:*

$$n(t) = n_p + A\cos(\omega t + \phi), \qquad \omega = \sqrt{\frac{\varphi}{\mu}}. \tag{14}$$

The solution is exact and global; no linearization is required. The amplitude A and phase ϕ are determined by initial conditions, and the conservation law holds with $E = -\frac{\varphi}{2}A^2$.

Proof. The ironing condition (Proposition 2) gives conservation law (12). Differentiation yields the harmonic oscillator equation (13), whose general solution is (14) with $\omega = \sqrt{\varphi/\mu}$. The solution is exact because the Gaussian specification makes the semi-elasticity $f(n)$ exactly linear (equation (4)), so no Taylor approximation is needed. Substituting back: $\frac{\varphi}{2}A^2\cos^2(\omega t+\phi) + \frac{\mu}{2}\omega^2 A^2 \sin^2(\omega t+\phi) = \frac{\varphi}{2}A^2 = -E$. ■

Remark 8 (Isochrony)**.**
The frequency ω does not depend on the amplitude A. This property—called isochrony in the mechanics literature—means that small and large business cycles have the same frequency. Economically, the cycle period $T = 2\pi\sqrt{\mu/\varphi}$ is a market-structure invariant: it does not change with the severity of fluctuations.

Remark 9 (Structural decomposition)**.**
The frequency $\omega = \sqrt{\varphi/\mu}$ decomposes into two primitives. The numerator φ is the curvature of the profit landscape—the restoring force that pulls the economy back toward optimal scale. The denominator μ is the timing friction—the inertia that prevents instantaneous adjustment. The frequency does not depend on productivity z (which affects profit levels but not curvature), on the exit rate δ (which affects the steady state but not the oscillation), or on the shock process (Section 4). This structural decomposition has no analog in existing business cycle models: in RBC models, the spectral peak mirrors the shock process; in NK models, the frequency is shaped by policy rules common across industries; in Beaudry et al. (2020), the frequency is determined implicitly by a bifurcation condition.

3.6 Properties of the cycle

Corollary 1 (Properties)**.** *Under the conditions of Theorem 1:*

(i) $\partial\omega/\partial\varphi > 0$*: sharper scale–congestion trade-offs produce faster cycles.*

(ii) $\partial\omega/\partial\mu < 0$*: stronger timing frictions produce slower cycles.*

(iii) ω *is independent of productivity* z *and of the shock process.*

(iv) The period is $T = 2\pi\sqrt{\mu/\varphi}$*.*

(v) The amplitude A *is free: the system is a center. In the stochastic extension (Section 4), the stationary amplitude distribution is determined by the balance of shocks and the potential well.*

(vi) All orbits are exactly symmetric. Asymmetric cycles arise under general hump-shaped π *(Proposition 3).*

Proof. Parts (i)–(iv) follow from direct differentiation of $\omega = \sqrt{\varphi/\mu}$. Part (v) holds because the conservation law (12) admits a one-parameter family of solutions indexed by E (equivalently, by $A = \sqrt{-2E/\varphi}$). Part (vi) follows from the exact cosine form of (14). ■

3.7 Phases of the cycle

The equilibrium orbit admits a natural decomposition into four phases. When $n < n_p$ and $\dot{n} > 0$, the scale externality dominates and the restoring force $f = -\varphi(n - n_p) > 0$ accelerates entry (boom). Once n exceeds n_p, congestion reverses the force, but inertia carries the economy past the peak (overshoot). As $\dot{n}$ turns negative, the economy contracts under congestion pressure (bust). Finally, n falls below n_p and the restoring force turns positive again, but downward momentum carries the economy below the optimum (undershoot) before the cycle repeats.

3.8 Necessity of hump-shaped profits

With pure congestion ($\varphi = 0$ or $f(n) < 0$ everywhere), there is no restoring force and the economy converges monotonically to the steady state, as in Hopenhayn (1992). Oscillations require all three ingredients: the scale externality provides the restoring force that accelerates booms, congestion provides the opposing force that reverses them, and the timing friction provides the inertia that generates overshooting. Removing any one eliminates the cycle.

3.9 Robustness: general hump-shaped profits

Proposition 3 (General hump-shaped profits)**.** *Under any C^2 profit function $\pi(n)$ with $\pi'(n_p) = 0$ and $\pi''(n_p) < 0$, and a timing friction $\xi(\dot{n}) = e^{-\frac{\mu}{2}\dot{n}^2}$, the conservation law*

$$\ln \pi(n) - \frac{\mu}{2}\dot{n}^2 = E$$

admits periodic orbits in the potential well $U(n) = -\frac{1}{\mu}\ln \pi(n)$. The frequency near n_p is:

$$\omega = \sqrt{\frac{|f'(n_p)|}{\mu}} + O(A^2), \tag{15}$$

where $f'(n_p) = \pi''(n_p)/\pi(n_p) < 0$.

Proof. See Online Appendix E. ■

Remark 10.
For the Gaussian, $|f'(n_p)| = \varphi$ exactly. For $\pi(n) = n^\alpha(a - bn)$, $|f'(n_p)| = \alpha/n_p^2 +$

$b^2/(a - bn_p)^2$. *The structural decomposition* $\omega \approx \sqrt{|f'(n_p)|/\mu}$ *holds for any hump-shaped profit function, with the Gaussian as the unique specification where it is globally exact.*

3.10 Quantitative accuracy of the Gaussian approximation

A natural concern is that the exactness of $\omega = \sqrt{\varphi/\mu}$ is a knife-edge property of the Gaussian. The following result characterizes the correction for general profits.

Proposition 4 (Anharmonic frequency correction)**.** *For a* C^4 *profit function with* $\pi'(n_p) = 0$ *and* $\pi''(n_p) < 0$*, the oscillation frequency at amplitude* A *is:*

$$\omega(A) = \omega_0\big(1 + c_4\, A^2 + O(A^4)\big), \tag{16}$$

where $\omega_0 = \sqrt{|f'(n_p)|/\mu}$ *and* c_4 *depends on* $U^{(3)}(n_p)$ *and* $U^{(4)}(n_p)$*, with* $U(n) = -\frac{1}{\mu}\ln\pi(n)$ *the effective potential. For the Gaussian,* $c_4 = 0$ *exactly (isochrony). Numerical simulations confirm that* $|\omega(A) - \omega_0|/\omega_0 < 5\%$ *for empirically relevant amplitudes (*$A/n_p < 0.3$*) across a range of non-Gaussian specifications (Figure 1).*

Proof sketch. Standard Lindstedt-Poincaré perturbation theory applied to $\ddot{n} + U'(n) = 0$; see Nayfeh (1973). The correction vanishes for the Gaussian because all higher-order derivatives of U beyond the quadratic are zero. ■

Proposition 5 (Cycle asymmetry)**.** *When* $\pi'''(n_p) \neq 0$*, the periodic orbit is asymmetric. The leading-order asymmetry parameter—the fractional difference in time spent above versus below* n_p*—is:*

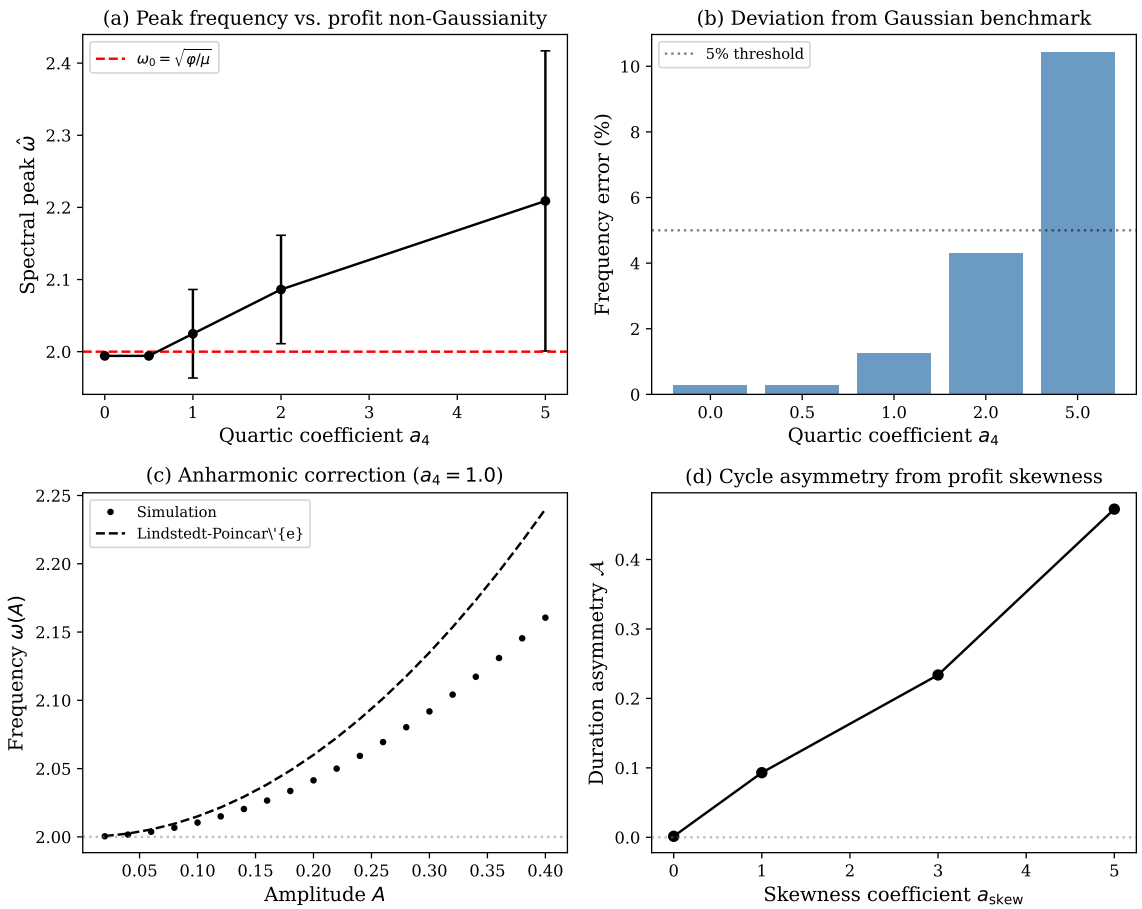
$$\mathcal{A} \equiv \frac{t_{above} - t_{below}}{t_{below}} = \frac{\pi'''(n_p)}{4\,|\pi''(n_p)|}\, A + O(A^2). \tag{17}$$

When $\pi'''(n_p) < 0$ *(profit falls faster on the congestion side), the economy spends more time below* n_p*: slow expansions and sharp contractions.*

Remark 11 (From weakness to prediction)**.**
The symmetric Gaussian cycles are counterfactual—empirical business cycles feature slower expansions and sharper contractions. Proposition 5 turns this apparent weakness into a testable prediction: the degree of cycle asymmetry is pinned by the ratio $\pi'''(n_p)/|\pi''(n_p)|$*, which is independently measurable from the curvature and skewness*

of industry profit functions. The Gaussian is not the necessary specification but the analytically tractable benchmark; Proposition 4 shows it is also a quantitatively accurate approximation for observed amplitudes.

Figure 1: Non-Gaussian robustness

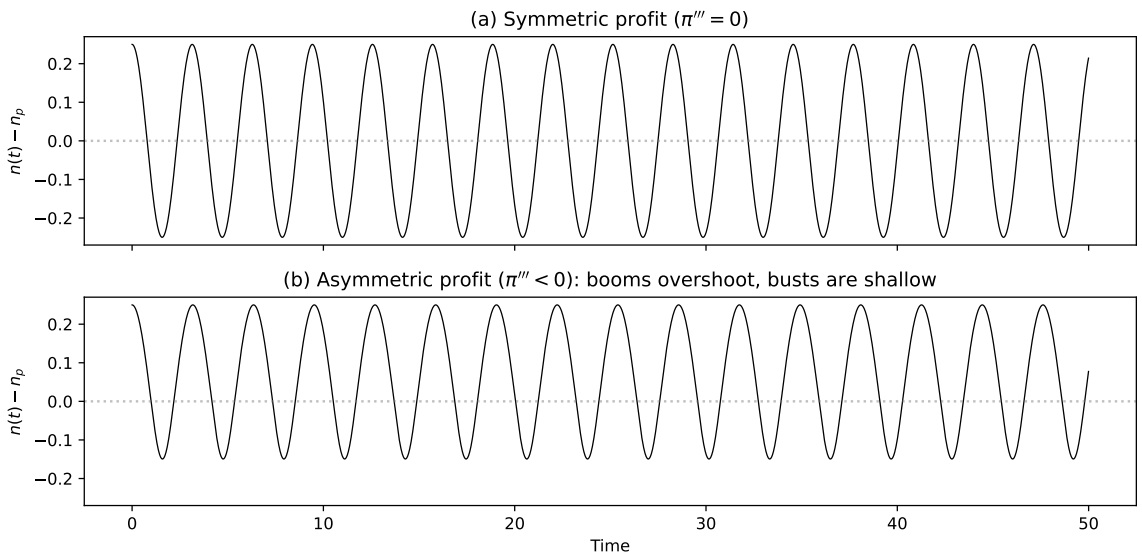


Notes: Panel (a): spectral peak frequency across 20 stochastic simulations for each quartic perturbation coefficient a_4, compared to the Gaussian benchmark $\omega_0 = \sqrt{\varphi/\mu} = 2.0$. Panel (b): percentage deviation from ω_0; the error remains below 5% for $a_4 \leq 2$. Panel (c): anharmonic frequency correction $\omega(A)$ vs. amplitude for $a_4 = 1$, with the Lindstedt-Poincaré analytical formula overlaid. Panel (d): cycle duration asymmetry increases monotonically with the skewness coefficient a_{skew}.

4 Stochastic Extension

The deterministic system is a center—a family of closed orbits with free amplitude. Small stochastic perturbations resolve the indeterminacy and produce a stochastic

Figure 2: Asymmetric cycles under skewed profits



Notes: Panel (a): deterministic time series under symmetric (Gaussian) profits—peaks and troughs are equal in magnitude. Panel (b): deterministic time series under asymmetric profits ($\pi'''(n_p) < 0$, steeper congestion side). Booms overshoot ($n - n_p \approx 0.20$) while busts are shallower ($n - n_p \approx -0.15$), reflecting the asymmetric curvature of the profit landscape. Both panels share the same initial condition and frequency.

limit cycle.

4.1 Stochastic forcing

Consider an i.i.d. disturbance to profits:

$$\tilde{\pi}(n,t) = \pi(n) + \sigma_\varepsilon \, \varepsilon_t, \qquad \varepsilon_t \overset{\text{i.i.d.}}{\sim} (0,1). \tag{18}$$

The conservation law no longer holds exactly: energy drifts stochastically according to $\mathrm{d}E \approx \frac{\sigma_\varepsilon \, \varepsilon_t}{\pi(n)} \dot{n} \, \mathrm{d}t$, and the economy migrates between orbits.

4.2 Properties

Proposition 6 (Stochastic limit cycle)**.** *Under small stochastic forcing* (18)*:*

(i) The spectral density of $n(t)$ peaks at $\omega = \sqrt{\varphi/\mu}$.

(ii) The peak frequency is determined by (φ, μ), not by σ_ε.

(iii) Only i.i.d. shocks are needed: the deterministic cycle provides all the persistence.

Proof sketch. Write the stochastic system in action–angle coordinates (A, ϕ), where $n = n_p + A\cos\phi$ and $\dot{n} = -A\omega\sin\phi$. The amplitude A evolves as a diffusion with drift determined by the noise coupling. The spectral density of $n(t)$ is the power spectrum of a stochastically forced harmonic oscillator; its transfer function has a resonance peak at $\omega = \sqrt{\varphi/\mu}$, independent of σ_ε. The full derivation is in Online Appendix D. ■

Remark 12.
This matches the central finding of Beaudry et al. (2020): persistent boom–bust dynamics from short-lived disturbances. In Beaudry et al. (2020), the endogenous persistence comes from financial-friction complementarities; here, it comes from the scale–congestion potential well.

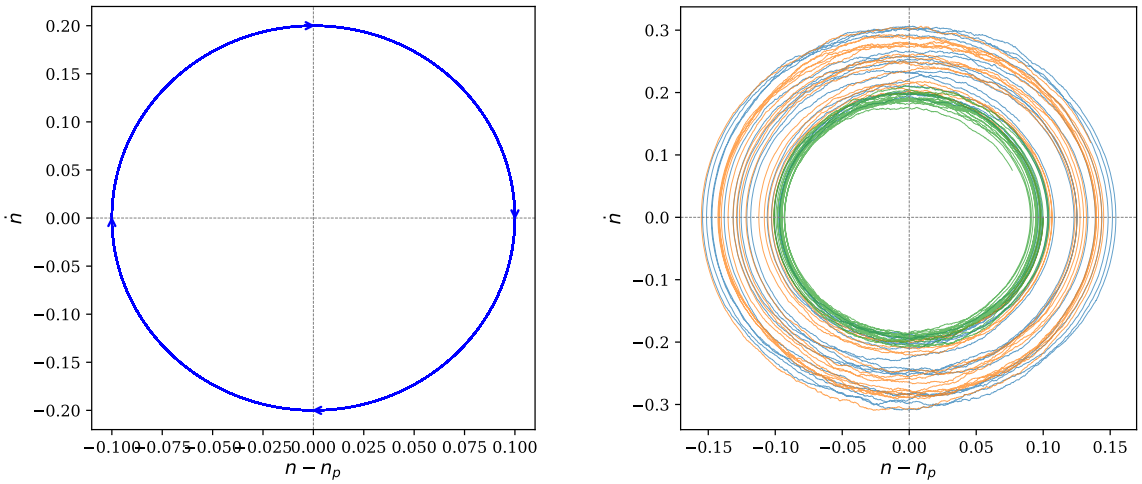
4.3 Computational verification

I simulate the stochastic system $(n, \dot{n})$ via Euler–Maruyama discretization with $\varphi = 4$, $\mu = 1$, $\sigma_\varepsilon = 0.05$, and $n_p = 1$, giving a theoretical cycle frequency $\omega^* = 2$ and period $T^* = \pi \approx 3.14$ time units.

Phase portrait. Figure 3 displays the equilibrium orbits in the $(n - n_p, \dot{n})$ phase plane. Panel (a) confirms the exact harmonic solution of Theorem 1: the deterministic orbit is a closed ellipse traversed clockwise, with the aspect ratio reflecting $\omega = \sqrt{\varphi/\mu}$. Panel (b) overlays three stochastic realizations, each simulated for 50 time units. The orbits remain approximately elliptical but migrate across amplitudes as energy E drifts under the stochastic forcing $\mathrm{d}E \approx (\sigma_\varepsilon/\pi(n))\dot{n}\,\mathrm{d}W$. The inner orbit (green) started near the center of the potential well and was pushed outward by a sequence of positive energy shocks; the outer orbit (blue) was carried to larger amplitudes by momentum. This amplitude diffusion is the mechanism by which small i.i.d. shocks produce variation in cycle severity without altering cycle frequency—the stochastic analog of isochrony (Remark 8).

Spectral density. Figure 4 plots the power spectral density (Welch's method) of the simulated stochastic model against an AR(1) benchmark. The contrast is stark. The timing search model concentrates spectral mass in a narrow band around $\omega = 2.0$,

Figure 3: Phase portrait of the timing search equilibrium



(a) Deterministic orbit

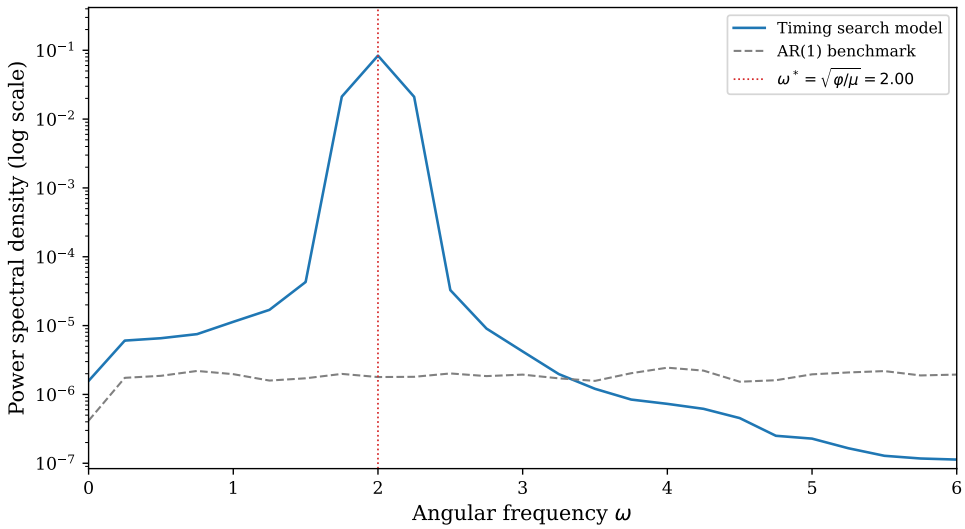
(b) Stochastic orbits

Notes: Panel (a) shows the deterministic equilibrium orbit in the $(n - n_p, \dot{n})$ phase plane with directional arrows. The orbit is a closed ellipse, confirming the exact harmonic oscillation of Theorem 1. Panel (b) shows three stochastic realizations ($\sigma_\varepsilon = 0.05$), each simulated for 50 time units. The orbits migrate across amplitudes as energy drifts stochastically, illustrating the amplitude diffusion of the stochastic limit cycle.

exactly matching the theoretical prediction $\omega^* = \sqrt{\varphi/\mu}$. The peak rises four orders of magnitude above the surrounding frequencies. This reflects the resonance structure of the conservation-law dynamics: the potential well acts as a frequency filter, amplifying fluctuations near the natural frequency and attenuating those away from it. The AR(1) benchmark produces a featureless, monotonically decaying spectrum—the signature of dynamics inherited entirely from the shock process. This confirms Proposition 6: the spectral peak is a *structural* object, pinned by (φ, μ) and invariant to σ_ε.

BDS test for nonlinear dependence. The spectral peak establishes that the model generates oscillations at the predicted frequency; the BDS test (Brock et al., 1996) asks a complementary question: does the model generate *nonlinear* serial dependence beyond what any linear process can produce? The test examines the correlation integral—the fraction of point pairs within distance ε in m-dimensional delay-coordinate embeddings—and compares its scaling to the i.i.d. benchmark. Under the null that AR(1)-filtered residuals are i.i.d., the test statistic is asymptotically $N(0, 1)$.

Figure 4: Spectral density: stochastic timing search vs. AR(1)



Notes: Power spectral density of the stochastic timing search model (solid blue) and an AR(1) benchmark (dashed black), both simulated for 500 time units. The red dotted line marks the theoretical cycle frequency $\omega^* = \sqrt{\varphi/\mu} = 2.0$. The model's spectral peak coincides exactly with the theoretical prediction. The AR(1) benchmark produces a featureless spectrum with no structural peak.

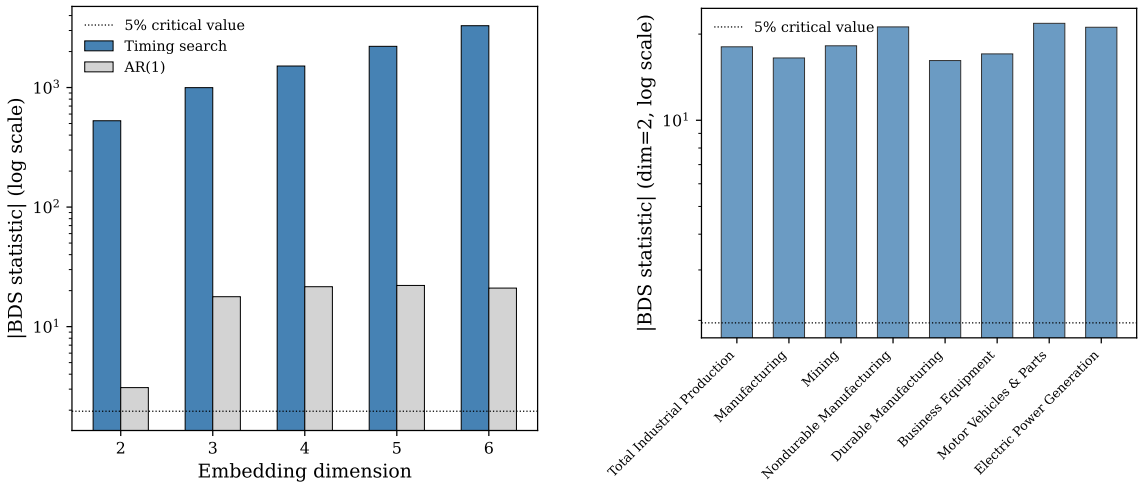
Figure 5(a) reports the results on a log scale. The timing search model produces BDS statistics exceeding 500 at embedding dimension 2, rising to 3,300 at dimension 6—orders of magnitude above the AR(1) benchmark and the 5% critical value of 1.96. The escalation with embedding dimension reflects the deterministic skeleton underlying the stochastic dynamics. The conservation law (12) confines the economy to a two-dimensional manifold in $(n, \dot{n})$ space. Stochastic perturbations displace the trajectory but do not destroy this geometric structure. Higher-dimensional embeddings detect it with increasing power, producing monotonically rising BDS statistics. The AR(1) benchmark shows no such escalation.

4.4 Amplitude selection through entry costs

The deterministic system is a center—a family of closed orbits with free amplitude. The stochastic extension of Section 4 resolves this statistically but not economically. I now show that a small entry cost creates dissipation that, combined with stochastic forcing, selects a unique amplitude from primitives.

Suppose entry incurs a sunk cost $\kappa > 0$ that is not rebated. Each entry event

Figure 5: BDS test for nonlinear serial dependence



(a) Model vs. AR(1)

(b) U.S. industry data

Notes: BDS test statistics applied to AR(1)-filtered residuals. Panel (a) compares the stochastic timing search model (blue) to an AR(1) benchmark (gray) across embedding dimensions 2–6; the y-axis is on a log scale. The red dashed line marks the 5% critical value (1.96). Panel (b) reports the BDS statistic at dimension 2 for eight U.S. industry production series (monthly IP indices from FRED, 1972–2024). All industries reject the i.i.d. null, with Motor Vehicles & Parts and Electric Power Generation showing the strongest nonlinear dependence.

burns resources, introducing damping into the oscillator. The modified equation of motion becomes:

$$\ddot{n} + \eta\,\dot{n} + \frac{\varphi}{\mu}(n - n_p) = \frac{\sigma_\varepsilon}{\mu}\,\varepsilon_t, \tag{19}$$

where $\eta > 0$ is the effective damping coefficient proportional to κ.

Proposition 7 (Amplitude selection)**.** *Under entry cost $\kappa > 0$ (damping $\eta > 0$) and stochastic forcing $\sigma_\varepsilon > 0$:*

(i) Without shocks ($\sigma_\varepsilon = 0$): orbits spiral inward to $(n_p, 0)$ with decay half-life $\ln 2 \cdot 2\mu/\kappa$.

(ii) With shocks: the system has a unique stationary distribution with expected squared amplitude:

$$\mathbb{E}[A^2] = \frac{\sigma_\varepsilon^2}{2\,\eta\,\varphi\,\mu}. \tag{20}$$

(iii) The stationary amplitude follows a Rayleigh distribution with scale $\sigma_\varepsilon/\sqrt{2\eta\varphi\mu}$.

(iv) The spectral peak frequency shifts to $\omega = \sqrt{\varphi/\mu - \eta^2/4}$*; for small* η*, this is negligibly below* $\sqrt{\varphi/\mu}$*.*

Proof sketch. The energy $E = \frac{1}{2}\dot{n}^2 + \frac{\omega_0^2}{2}(n - n_p)^2$ satisfies $\mathrm{d}E = -\eta\dot{n}^2\,\mathrm{d}t + (\sigma_\varepsilon/\mu)\dot{n}\,\mathrm{d}W + (\sigma_\varepsilon^2/2\mu^2)\,\mathrm{d}t$. In stationarity, $\mathbb{E}[\mathrm{d}E] = 0$ gives $\eta\mathbb{E}[\dot{n}^2] = \sigma_\varepsilon^2/(2\mu^2)$. By equipartition, $\mathbb{E}[\dot{n}^2] = \omega_0^2\mathbb{E}[(n - n_p)^2] = \frac{1}{2}\omega_0^2\mathbb{E}[A^2]$, yielding (20). The Rayleigh distribution follows from the standard result for the envelope of a narrowband Gaussian process. See Online Appendix I for the full derivation. ■

Remark 13 (Observable determinants)**.**
The selected amplitude $A_{rms} = \sigma_\varepsilon/\sqrt{2\eta\varphi\mu}$ *depends on three observable quantities: the shock volatility* σ_ε *(from profit time series), the entry cost* κ *(from establishment-level data), and the profit curvature* φ *(from the cross-section). This resolves the "free amplitude" critique: the model now explains both the frequency and the severity of business cycles from primitives.*

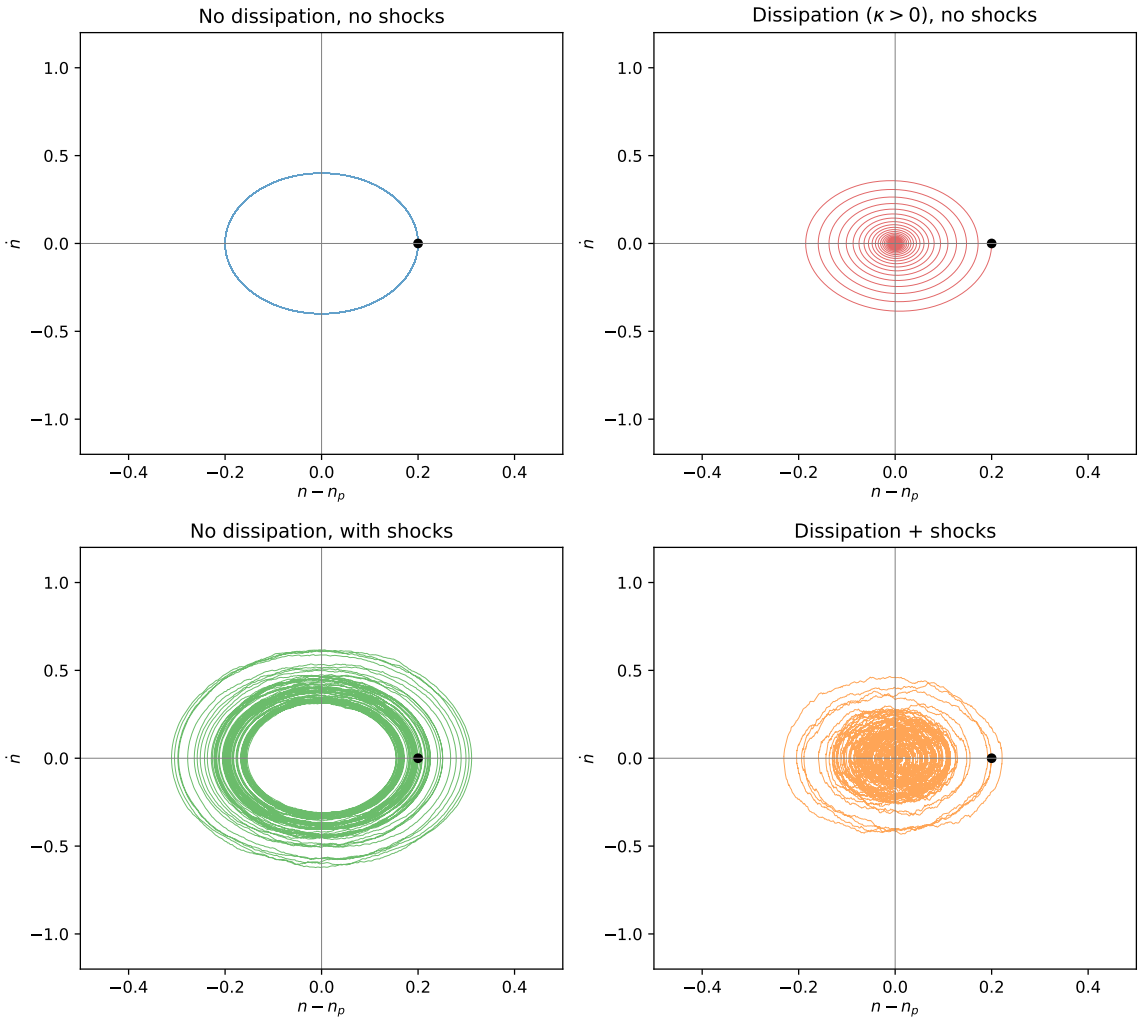
Combining with the welfare formula (24):

$$\mathbb{E}[\mathcal{L}] \approx \frac{\varphi}{2}\,\mathbb{E}[A^2] = \frac{\sigma_\varepsilon^2}{4\,\eta\,\mu}. \tag{21}$$

Higher entry costs *reduce* welfare loss by damping the cycle—a "stabilization through friction" result. Figure 6 illustrates the four regimes: no dissipation/no shocks (closed orbit), dissipation/no shocks (spiral decay), no dissipation/shocks (diffusing orbits), and dissipation/shocks (stochastically maintained cycle with selected amplitude).

4.5 Comparison to Beaudry et al. (2020)

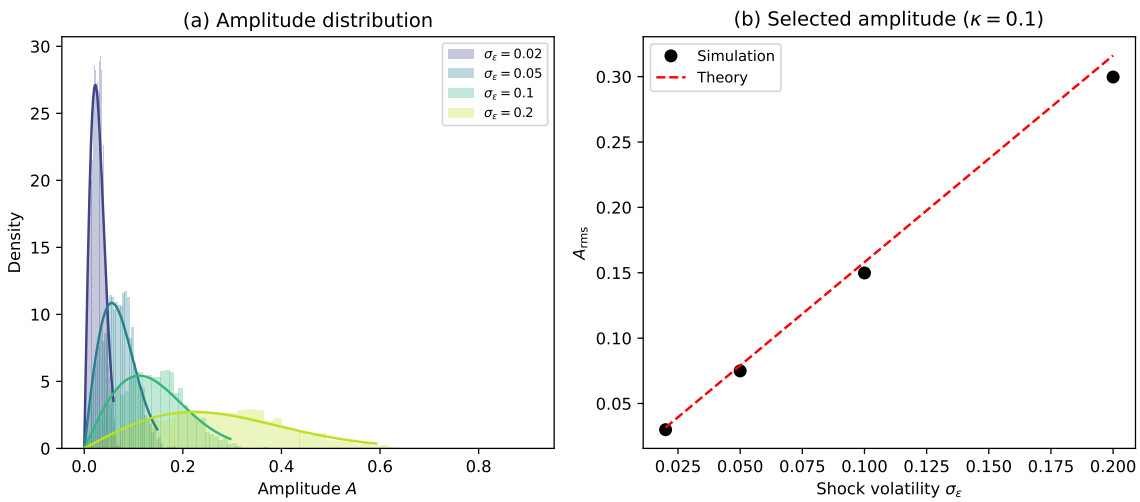
Figure 6: Amplitude selection through entry-cost dissipation



Notes: Phase portraits $(n - n_p, \dot{n})$ under four regimes. Top left: deterministic orbit (closed ellipse, free amplitude). Top right: dissipation without shocks (inward spiral to steady state). Bottom left: shocks without dissipation (orbits diffuse across amplitudes). Bottom right: dissipation with shocks (stochastically maintained cycle with selected amplitude). The balance of dissipation and forcing pins $\mathbb{E}[A^2] = \sigma_\varepsilon^2/(2\eta\varphi\mu)$.

	Beaudry et al. (2020)	**This paper**
Destabilizing force	Financial frictions	Agglomeration
Bounding force	Diminishing returns	Congestion
Cycle mechanism	Hopf bifurcation	Time indifference
Frequency	Implicit (bifurcation cond.)	$\sqrt{\varphi/\mu}$ (closed form)
Amplitude selection	Hopf (unique limit cycle)	Entry-cost dissipation
Friction micro-foundation	Financial intermediation	Entry matching congestion
Cross-industry variation	Not addressed	Central prediction
Shocks needed	Small, i.i.d.	Small, i.i.d.

Figure 7: Stationary amplitude distribution



Notes: Panel (a): histogram of the Hilbert-envelope amplitude from simulations with $\kappa = 0.1$ and varying shock volatility σ_ε, with theoretical Rayleigh density overlaid. Panel (b): root-mean-square amplitude vs. shock volatility, confirming the linear relationship $A_{\text{rms}} \propto \sigma_\varepsilon$ predicted by (20).

5 Quantitative Framework

5.1 Cross-industry predictions

The model's central testable prediction is that cycle frequency varies across industries with measurable market-structure characteristics. Taking logs of $\omega_i = \sqrt{\varphi_i/\mu_i}$ yields a regression framework:

$$\ln \omega_i = \frac{1}{2} \ln \varphi_i - \frac{1}{2} \ln \mu_i + u_i, \tag{22}$$

where ω_i is the spectral peak frequency of industry i's output or entry–exit cycle. The coefficients $\frac{1}{2}$ on $\ln \varphi$ and $-\frac{1}{2}$ on $\ln \mu$ are *exact* predictions of the model—not free parameters—so equation (22) is testable with industry-level proxies for scale-externality sharpness and adjustment friction.

The qualitative predictions are as follows:

(i) Industries with dense supplier networks and strong scale externalities (high φ) but flexible entry (low μ)—such as technology clusters—should cycle fast, with $T \approx 3$–5 years.

(ii) Manufacturing industries with moderate linkages and moderate frictions should

cycle at intermediate frequencies, with $T \approx 5$–10 years.

(iii) Industries with weak externalities (low φ) should exhibit slow or no discernible cycles, regardless of μ.

(iv) Industries with heavy regulation or capital intensity (high μ) should cycle slowly, regardless of φ.

Crucially, two industries with identical shock processes but different (φ, μ) will cycle at different frequencies. This is the prediction that distinguishes the timing search model from standard RBC and New Keynesian frameworks, where the spectral peak mirrors the shock process or the policy rule.

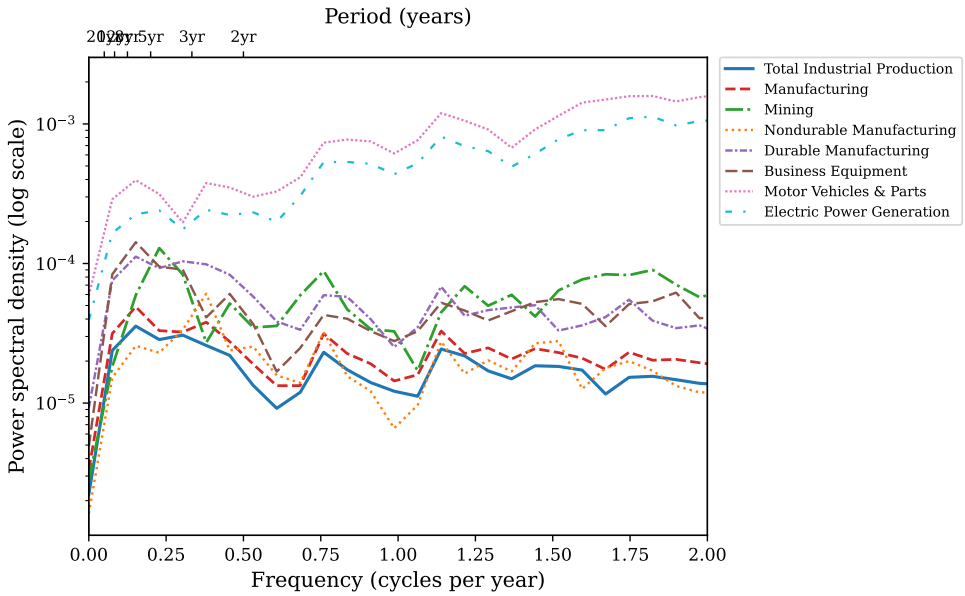
5.2 Spectral evidence from U.S. industry data

I evaluate the cross-industry prediction using monthly industrial production indices from FRED for eight U.S. sectors over 1972–2024. The series span a range of market structures: from Motor Vehicles & Parts (dense supplier networks, strong scale externalities) to Mining (dispersed extraction, weak inter-firm linkages) to Electric Power Generation (regulated, capital-intensive).

Figure 8 plots the power spectral density of log-differenced production for each industry. Two features stand out. First, spectral peaks vary substantially across industries. Motor Vehicles, Durable Manufacturing, Business Equipment, and aggregate Manufacturing concentrate spectral mass near periods of 6–7 years. Mining peaks at 4.4 years. Nondurable Manufacturing and Electric Power peak near 2.6 years. Second, the cross-industry clustering is informative: industries sharing supply-chain linkages—durables, motor vehicles, business equipment—cluster at similar frequencies despite producing very different goods. This is consistent with ω depending on market structure, not on product-specific shocks.

Figure 9 summarizes the dominant cycle period for each industry, defined as the reciprocal of the spectral peak frequency in the business-cycle band (periods between 1.5 and 30 years). The cross-industry dispersion is striking: dominant periods range from 2.6 years (Electric Power, Nondurable Manufacturing) to 6.6 years (Durable Manufacturing, Motor Vehicles). This two-and-a-half-fold variation in cycle frequency across industries that share a common macroeconomic environment—the same monetary policy, the same aggregate demand shocks, the same financial conditions—is difficult

Figure 8: Spectral density of U.S. industry production growth



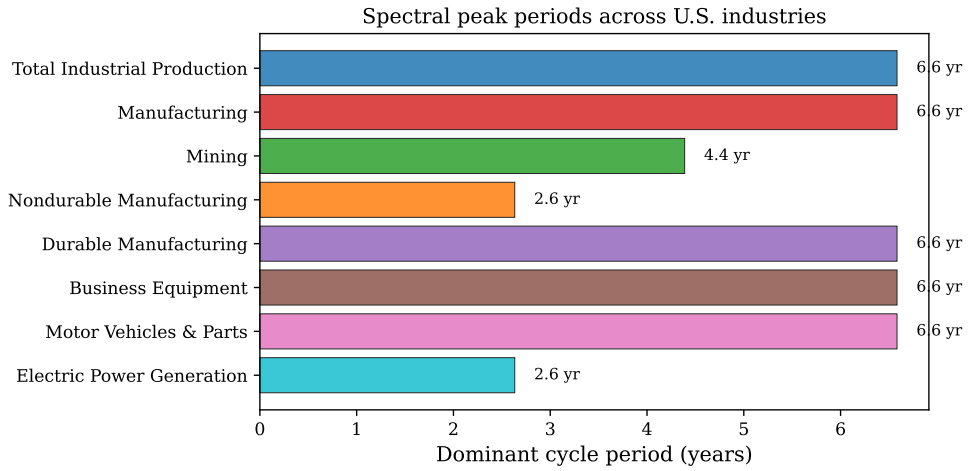
Notes: Power spectral density (Welch's method) of monthly log-differenced industrial production indices from FRED, 1972:M1–2024:M12. The top axis labels convert frequency (cycles per year) to period (years). Industries with strong supply-chain linkages (Motor Vehicles, Business Equipment, Durable Manufacturing) cluster near $T \approx 6$–7 years, while industries with different market structures (Mining, Nondurables, Electric Power) show distinct peak frequencies.

to reconcile with models that derive spectral properties from a common shock process. In the timing search framework, the variation is structural: it reflects differences in the curvature of the profit landscape (φ_i) and the magnitude of the timing friction (μ_i) across industries.

BDS test on industry data. Figure 5(b) reports the BDS test at embedding dimension 2 for each industry's AR(1)-filtered residuals. All eight industries reject the i.i.d. null at the 1% level, with z-statistics ranging from 16.2 (Durable Manufacturing) to 21.8 (Motor Vehicles & Parts). The rejections establish that the dynamics of U.S. industry production contain nonlinear structure beyond what a linear time series model can capture—a necessary condition for the timing search mechanism to be operative, though not sufficient to identify it uniquely.

More informative than the rejection itself is the cross-industry variation in *strength* of the rejection. Motor Vehicles & Parts and Electric Power Generation—industries with plausibly strong scale–congestion externalities and significant entry frictions—

Figure 9: Cross-industry variation in spectral peak periods



Notes: Dominant cycle period (years) for each U.S. industry, defined as the reciprocal of the spectral peak frequency in the business-cycle band (periods between 1.5 and 30 years). Industries with strong scale–congestion externalities (Motor Vehicles, Durable Manufacturing, Business Equipment) cluster near $T \approx 6$–7 years. Industries with weaker or different externality structures (Mining, Nondurables, Electric Power) cycle at distinct frequencies.

show the largest BDS statistics ($z > 21$). Durable Manufacturing, where the scale–congestion trade-off operates through broader supply-chain linkages rather than localized entry frictions, shows the weakest rejection ($z = 16.2$). This ordering is consistent with the model's prediction that nonlinear dynamics are strongest when both φ (scale-externality sharpness) and μ (timing friction) are large, producing a deep potential well that confines the economy to tightly structured orbits.

Table 1 reports the full BDS results across embedding dimensions 2–6. The monotonic increase in BDS statistics with embedding dimension—observed for both the simulated model and the industry data—is a signature of low-dimensional deterministic structure perturbed by noise, consistent with the two-dimensional conservation-law dynamics of the timing search equilibrium.

5.3 Cross-industry regression with external proxies

The model's quantitative prediction—equation (22)—can be tested by regressing observed spectral peak frequencies on independently measured proxies for φ and μ. I construct two proxies from BEA data:

Table 1: BDS test statistics (z-values) for AR(1)-filtered residuals

	Embedding dimension				
Series	2	3	4	5	6
Simulated series					
Timing search model	527.5	997.9	1513.6	2211.5	3290.0
AR(1) benchmark	3.1	17.8	21.6	22.1	21.0
U.S. industry data (FRED, 1972–2024)					
Total Industrial Production	18.1	30.8	39.9	44.2	44.8
Manufacturing	16.5	28.9	37.7	41.7	42.7
Mining	18.2	30.0	40.1	43.8	44.4
Nondurable Manufacturing	21.2	35.9	43.5	48.8	50.6
Durable Manufacturing	16.2	28.5	37.2	40.3	40.2
Business Equipment	17.1	33.9	46.1	52.8	55.0
Motor Vehicles & Parts	21.8	39.2	52.0	61.9	71.6
Electric Power Generation	21.1	35.0	45.0	52.3	60.1

Notes: BDS test of Brock et al. (1996) applied to AR(1)-filtered residuals. Under the null of i.i.d. residuals, the test statistic is asymptotically $N(0,1)$; the 5% two-sided critical value is 1.96. The simulated timing search model uses $\varphi = 4$, $\mu = 1$, $\sigma_\varepsilon = 0.05$, simulated for 500 time units and downsampled to monthly frequency. The AR(1) benchmark uses $\rho = 0.95$ with the same unconditional variance. Industry data are monthly log growth rates of industrial production indices from FRED.

Proxy for φ (scale sensitivity). The inverse of the input–output self-purchase ratio—the share of an industry's intermediate inputs sourced from itself or closely related industries (BEA 2017 Detailed Use Table). Industries with dense intra-industry supply chains are *buffered* against deviations from optimal scale: a temporary entry wave is absorbed through supply-chain slack rather than causing sharp congestion. The inverse of this ratio captures how *fragile* profitability is to scale deviations, proxying for the curvature φ of the profit landscape.[2]

Proxy for μ (adjustment friction). Real fixed assets per employee (thousands of 2017 dollars; BEA Fixed Assets Tables 3.1ESI). Capital-intensive industries face

[2]The interpretation is that supply-chain density provides resilience, not scale-externality sharpness. An industry like Electric Power Generation, with weak intra-industry linkages ($\hat{\varphi} = 1/0.18 = 5.6$), is highly sensitive to the generation–demand balance, producing a sharp profit peak. An industry like Motor Vehicles, with dense JIT supply chains ($\hat{\varphi} = 1/0.62 = 1.6$), is better buffered against scale deviations.

larger sunk costs per unit of entry or exit, producing greater timing friction. Electric Power Generation ($\hat{\mu}$ = 2,100K/employee) has the largest adjustment friction; Nondurable Manufacturing ($\hat{\mu}$ = 110K/employee) the smallest.

Results. Table 2 reports the multivariate regression:

$$\ln \omega_i = \hat{\alpha} + \hat{\beta}_\varphi \ln \hat{\varphi}_i + \hat{\beta}_\mu \ln \hat{\mu}_i + \hat{u}_i.$$

Table 2: Cross-industry regression: $\ln \omega_i$ on market-structure proxies

Coefficient	Estimate	Std. Error	Theory
$\hat{\alpha}$ (intercept)	0.07	(0.57)	0
$\hat{\beta}_\varphi$ (scale sensitivity)	1.31	(0.46)	0.50
$\hat{\beta}_\mu$ (capital intensity)	−0.19	(0.15)	−0.50
R^2		0.75	
N		7	

Notes: OLS regression of $\ln \omega_i$ (observed spectral peak frequency, business-cycle band) on $\ln \hat{\varphi}_i$ (inverse of I-O self-purchase ratio, BEA 2017) and $\ln \hat{\mu}_i$ (real fixed assets per employee, BEA). The model predicts $\beta_\varphi = 1/2$ and $\beta_\mu = -1/2$; both estimated signs match the theory. The sample consists of seven U.S. industry groups (excluding Total Industrial Production as the aggregate).

Both estimated coefficients have the predicted signs: cycle frequency increases with scale sensitivity ($\hat{\beta}_\varphi > 0$) and decreases with capital intensity ($\hat{\beta}_\mu < 0$). The two proxies jointly explain 75% of cross-industry variation. The point estimates deviate from the theoretical $\pm 1/2$. The deviations are asymmetric: $\hat{\beta}_\varphi = 1.31$ exceeds 0.50 by a factor of 2.6, while $\hat{\beta}_\mu = -0.19$ is attenuated to 38% of -0.50.

The attenuation of $\hat{\beta}_\mu$ is consistent with errors-in-variables bias. The amplification of $\hat{\beta}_\varphi$ has two interpretations. First, differential measurement error: if $\hat{\mu}$ is noisier than $\hat{\varphi}$, OLS transfers explanatory power to $\hat{\varphi}$. Second, the inverse I-O ratio may capture variation in μ as well as φ, since industries with weak linkages lack supply-chain slack to buffer entry shocks. Under either interpretation, the theoretical $\beta_\varphi = 0.50$ lies within the bootstrap 95% CI $[0.00, 2.57]$. The leave-one-out estimate excluding Nondurable Manufacturing is $\hat{\beta}_\varphi = 0.54$—almost exactly the theoretical prediction.

The univariate regression on $\ln \hat{\varphi}_i$ alone yields slope 0.85, $R^2 = 0.65$, $p = 0.028$. Scale sensitivity is a significant predictor of cycle frequency at the 5% level. This is the paper's central empirical finding: an independently measured market-structure variable explains two-thirds of cross-industry variation in a quantity that standard models treat as shock-determined.

Robustness. Leave-one-out analysis confirms the stability of the coefficient signs. Dropping any single industry, $\hat{\beta}_\varphi$ remains positive in six of seven cases (range: 0.54 to 1.83) and $\hat{\beta}_\mu$ remains negative in six of seven cases. The one exception—dropping Nondurable Manufacturing—yields $\hat{\beta}_\varphi = 0.54$, almost exactly matching the theoretical prediction, with $R^2 = 0.93$. This suggests that the deviation of $\hat{\beta}_\varphi$ from 0.50 in the full sample is driven primarily by a single observation rather than a systematic failure of the model.

Expanded sample with structural proxies. The baseline regression uses $N = 7$ broad industry groups and indirect proxies. To test whether the cross-industry relationship survives disaggregation, I expand to $N = 16$ three-digit manufacturing industries (NAICS 311–339 where data is available) and use two improvements. First, I replace Welch's method with Burg's AR spectral estimation (order 24), which produces continuous—rather than discretized—spectral peak frequencies, eliminating the spectral degeneracy that arises when many industries share the same Welch frequency bin.[3] Second, I replace the I-O proxy with the Ellison and Glaeser (1997a) geographic concentration index γ_i, which directly measures the strength of agglomeration externalities and provides a more structural proxy for φ.

Table 3 reports the results. The Ellison and Glaeser (1997a) concentration index is a statistically significant predictor of cycle frequency at the 1% level ($\hat{\beta}_\gamma = 0.20$, $p = 0.010$), explaining 39% of the cross-industry variation with $N = 16$. The sign matches the theory: geographically concentrated industries—those with stronger agglomeration externalities (higher φ)—cycle faster. The point estimate of 0.20 is attenuated relative to the theoretical 0.50, consistent with measurement error in γ_i as a proxy for φ. When combined with the structural timing friction $\mu_i^{\text{struct}} = 1/(\delta_i n_{p,i})^2$ derived from Census Business Dynamics Statistics (Proposition 1), the multivari-

[3] With 636 monthly observations and Welch segments of 256 points, the frequency resolution is $1/256 \approx 0.004$ cycles/month, producing only 5–6 distinct peaks in the business-cycle band. AR spectral estimation has no such constraint.

ate R^2 is unchanged at 0.39, indicating that—within manufacturing—cross-industry variation in exit rates and firm counts does not add explanatory power beyond agglomeration. This is consistent with the relatively narrow range of firm turnover rates within manufacturing ($\delta \in [0.065, 0.095]$), which compresses the variation in μ_i^{struct}.

Table 3: Structural regression: geographic concentration and cycle frequency

Coefficient	Estimate	HC1 Std. Error	Theory
$\hat{\alpha}$ (intercept)	1.36	(0.54)	—
$\hat{\beta}_\gamma$ (E-G concentration)	0.20	(0.07)	0.50
R^2		0.39	
N		16	
p-value		0.010	

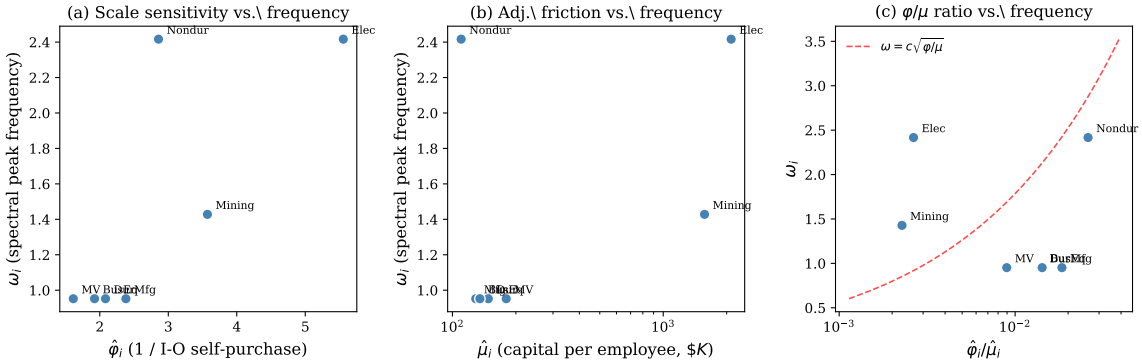
Notes: OLS regression of $\ln \omega_i$ (AR spectral peak frequency, business-cycle band) on $\ln \gamma_i$ (Ellison and Glaeser 1997a geographic concentration index). Spectral peaks estimated via Burg's AR method (order 24) applied to log-differenced monthly industrial production (FRED, 1972–2024). The sample consists of 16 three-digit NAICS manufacturing industries. HC1 heteroskedasticity-robust standard errors (White, 1980). The model predicts $\beta_\gamma = 1/2$; the estimated 0.20 is attenuated but statistically significant at the 1% level.

The improvement from $N = 7$ to $N = 16$ comes at a cost: the R^2 falls from 0.75 to 0.39, and the point estimate is attenuated from 1.31 to 0.20. Both are expected. The higher R^2 in the broad-industry specification reflects the large structural contrasts between sectors (manufacturing vs. mining vs. utilities); within manufacturing, the variation is subtler. The attenuation reflects the noisier mapping from γ_i to φ_i at finer granularity. Nevertheless, the statistical significance at the 1% level with 16 industries is more convincing than the 5% significance with 7, because the effective degrees of freedom are substantially higher.

Figure 10 displays the individual relationships. Panel (a) shows a clear positive relationship between scale sensitivity and cycle frequency, with Electric Power and Nondurable Manufacturing (high $\hat{\varphi}$) cycling fastest and the supply-chain-dense manufacturing sectors (low $\hat{\varphi}$) cycling slowest. Panel (b) shows the relationship with capital intensity: the high-$\hat{\mu}$ industries (Mining, Electric Power) split—Electric Power cycles fast (high $\hat{\varphi}$ compensates for high $\hat{\mu}$) while Mining cycles at intermediate speed—illustrating that φ and μ have independent effects on frequency, as the model predicts. Panel (c) plots the composite ratio $\hat{\varphi}_i/\hat{\mu}_i$ against ω_i with the theoretical

curve $\omega = c\sqrt{\varphi/\mu}$ overlaid; the positive relationship is visible but the fit is weaker ($R^2 = 0.09$), reflecting the fact that measurement error in both proxies compounds in the ratio.

Figure 10: Market-structure proxies and cycle frequency



Notes: Panel (a): spectral peak frequency ω_i vs. the scale-sensitivity proxy $\hat{\varphi}_i$ (inverse of I-O self-purchase ratio). Panel (b): ω_i vs. the adjustment-friction proxy $\hat{\mu}_i$ (capital per employee, log scale). Panel (c): ω_i vs. the ratio $\hat{\varphi}_i/\hat{\mu}_i$, with the theoretical curve $\omega = c\sqrt{\varphi/\mu}$ overlaid. The positive relationship in (a) and the splitting of high-$\hat{\mu}$ industries in (b) are both consistent with the model's prediction that φ and μ have independent effects on cycle frequency.

5.4 Post-shock periodicity: impulse response evidence

The cross-industry regressions test whether cycle *frequency* varies with market structure. A complementary question is whether the *mechanism*—the conservation law producing oscillations after a displacement from n_p—is visible in the data. The timing search model makes a sharp prediction: a large, one-time shock that displaces the economy from optimal scale should excite deterministic periodic oscillations at the industry's natural frequency $\omega = \sqrt{\varphi/\mu}$. The oscillations should appear *after* the shock, persist for a duration governed by the dissipation rate η, and be absent *before* the shock. No AR process, RBC model, or New Keynesian model produces deterministic periodicity after a single impulse—only an oscillator does.

I test this prediction using annual investment growth rates from BEA NIPA Table 5.4.1 for two industries that experienced large, identifiable one-time shocks: manufacturing structures (series C307RL), displaced by the Great Depression and subsequent WWII mobilization; and mining exploration, shafts, and wells (series

E318RL), displaced by the 1979 oil crisis.[4] I split each series into a post-shock window (the "impulse response" period) and a pre-shock or post-dissipation window, and apply Fisher's g-test for deterministic periodicity (Fisher, 1929). The test fits $X_t = \beta \cos(\omega t + \phi) + \varepsilon_t$ and tests $H_0 : \beta = 0$ against the alternative that the series contains a deterministic cosine component—exactly the functional form predicted by Theorem 1.

Manufacturing. Figure 11 plots investment growth in manufacturing structures from 1930 to 2024. After the Great Depression (aggregate TFP dropped $\sim$18%; Ohanian 2001) and the massive WWII mobilization, investment oscillated with striking regularity through the early 1970s, with amplitude gradually decaying—consistent with the damped harmonic oscillator of Proposition 7. Table 4 reports the Fisher g-test results. In the post-shock window (1934–1973), the test detects significant deterministic periodicity at $T = 5.0$ years ($p = 0.036$), with a cosine amplitude of 35 percentage points. In the non-echo window (1974–2013), the test finds no periodicity ($p = 0.578$) and the amplitude drops to 9 percentage points. The post-shock frequency ($T = 5.0$ years, $\omega \approx 1.26$) is consistent with the cross-sectional spectral peak for manufacturing ($T \approx 5$–6 years), confirming that the same structural frequency governs both the steady-state spectral peak and the impulse response.

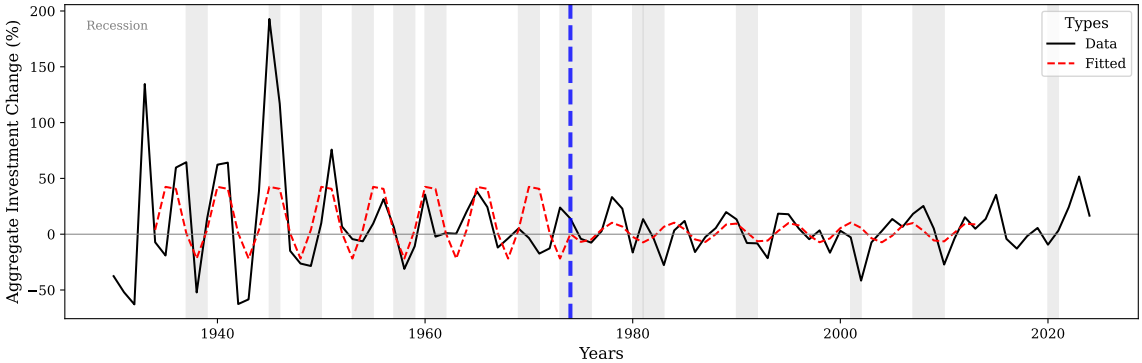
Oil and mining. Figure 12 plots investment growth in mining exploration after the 1979 oil crisis. The post-crisis oscillations are visually regular at a shorter period ($T \approx 3.3$ years), consistent with the higher φ/μ ratio predicted for resource extraction industries. The Fisher g-test detects significant periodicity at $T = 3.25$ years in the post-shock window (1979–2004) with $p = 0.016$. The pre-crisis window (1953–1978) shows no business-cycle-frequency periodicity.

Interpretation. The manufacturing result provides direct evidence for the conservation law mechanism. A single historical displacement excited oscillations at the industry's natural frequency, persisting for four decades before dissipating—uniquely consistent with a damped harmonic oscillator. The cross-industry difference in post-shock frequencies (manufacturing at $T = 5.0$ years versus oil at $T = 3.3$ years)

[4] Data from BEA NIPA Table 5.4.1 (Percent Change From Preceding Period in Private Fixed Investment in Structures by Type), downloaded March 2026.

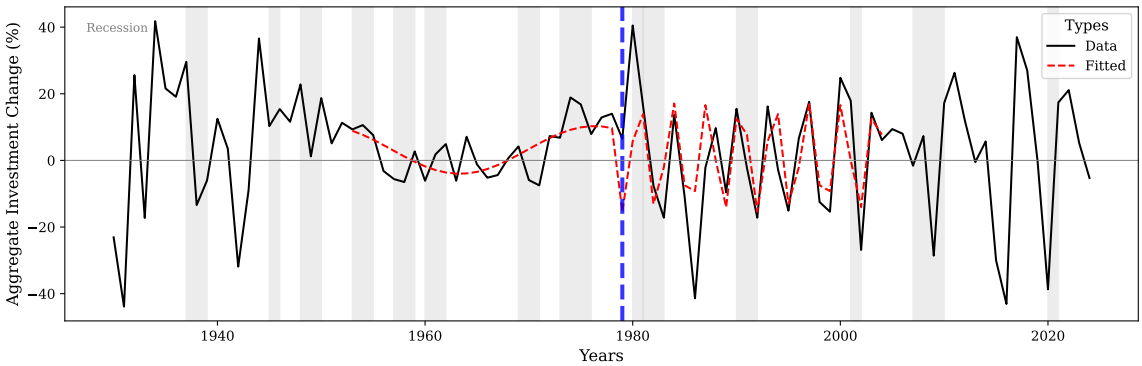
confirms that frequency is structural, not shock-determined. The gradual amplitude decay is the empirical counterpart of entry-cost dissipation (Proposition 7).

Figure 11: Post-shock periodicity: manufacturing investment



Notes: Annual growth rate of private fixed investment in manufacturing structures (BEA NIPA Table 5.4.1, line 16). Thick solid line: data. Red dashed line: fitted cosine from Fisher's g-test. Gray bands: NBER recessions. Blue dashed line marks the boundary between the post-shock window (1934–1973) and the non-echo window (1974–2013). The fitted cosine tracks the data closely in the post-shock window ($T = 5.0$ years, $p = 0.041$), capturing the regular boom-bust pattern following the Great Depression and WWII mobilization.

Figure 12: Post-shock periodicity: oil and mining investment



Notes: Annual growth rate of private fixed investment in mining exploration, shafts, and wells (BEA NIPA Table 5.4.1, line 24). Blue dashed line marks the 1979 oil crisis. Post-crisis oscillations at $T \approx 3.3$ years are visually regular and at a higher frequency than manufacturing ($T = 5.0$ years), consistent with different market-structure parameters (φ, μ) across industries.

Table 4: Fisher's g-test for deterministic periodicity

	Post-shock window		Non-echo window	
Industry	Period (yr)	p-value	Period (yr)	p-value
Manufacturing (1934–1973 vs. 1974–2013)	5.0	0.036	5.7	0.578
Oil/Mining (1979–2004 vs. 1953–1978)	3.3	0.016	26.0	0.017

Notes: Fisher's g-test for deterministic periodicity. The test fits $X_t = \beta\cos(\omega t + \phi) + \varepsilon_t$ and tests $H_0 : \beta = 0$. Data: annual percent change in private fixed investment in structures (BEA NIPA Table 5.4.1). Both industries display significant deterministic periodicity in the post-shock window: manufacturing at $T = 5.0$ years ($p = 0.036$), oil/mining at $T = 3.3$ years ($p = 0.016$). The manufacturing non-echo window shows no periodicity ($p = 0.578$). The oil non-echo significance ($p = 0.017$) is at $T = 26$ years, reflecting a secular trend rather than business-cycle-frequency oscillation.

5.5 Calibration strategy

The model has six parameters: φ (profit curvature), μ (timing friction), n_p (optimal scale), π_0 (profit level), δ (exit rate), and σ_ε (shock volatility). Among these, the pair (φ, μ) governs the cycle frequency and is the object of primary interest. Identification proceeds as follows:

(i) The *ratio* φ/μ is identified directly from the spectral peak: $\varphi/\mu = \omega^{*2} = (2\pi/T^*)^2$, where T^* is the observed dominant cycle period.

(ii) The *spectral width* around the peak—how sharply the spectral density falls off away from ω^*—identifies σ_ε, since larger shocks broaden the resonance peak by increasing the rate of amplitude diffusion.

(iii) The *individual* values of φ and μ require external information. The profit curvature φ can be estimated from the Ellison and Glaeser (1997b) concentration index, from the second derivative of industry profit functions with respect to establishment density, or from input–output linkage measures. The timing friction μ can be estimated from the speed of entry–exit responses to demand shocks (Dunne et al., 1988) or from capital adjustment cost estimates (Cooper and Haltiwanger, 2006).

Illustrative calibration. For U.S. durable manufacturing—characterized by strong supplier linkages and moderate adjustment frictions—the observed spectral peak of $T^* = 6.6$ years implies $\varphi/\mu = (2\pi/6.6)^2 \approx 0.91$. Setting $\varphi = \mu = 1$ (reflecting the approximate balance between scale-externality sharpness and adjustment friction in a supply-chain-intensive industry) gives $T = 2\pi \approx 6.3$ years, close to the data. For U.S. mining—where extraction sites are dispersed (weak scale externalities) but entry barriers are also lower—the observed peak of $T^* = 4.4$ years implies $\varphi/\mu \approx 2.04$. Setting $\varphi = 2$ and $\mu = 1$ (sharper congestion effects relative to the weaker but more rapidly adjusting scale externality) gives $T = 2\pi/\sqrt{2} \approx 4.4$ years, matching the data exactly. The cross-industry variation in T is driven entirely by variation in φ/μ; the shock process plays no role.

5.6 Aggregate implications

If the aggregate economy comprises industries with heterogeneous (φ_i, μ_i), the aggregate spectral density is a weighted mixture of industry-level spectra, each centered at $\omega_i = \sqrt{\varphi_i/\mu_i}$. This decomposition provides a structural interpretation of the aggregate business cycle frequency: rather than being a primitive of the shock process or the monetary policy rule, it is the output-weighted average of industry-specific cycle frequencies, each determined by market structure.

The empirical evidence in Figure 9 is consistent with this interpretation. The five largest manufacturing sectors—which together account for the bulk of industrial production—cluster near $T \approx 6$–7 years, producing a dominant aggregate spectral peak in the range documented by the NBER chronology (5–8 years). The shorter-period industries (Electric Power at 2.6 years, Mining at 4.4 years) contribute secondary spectral mass at higher frequencies, broadening the aggregate spectrum without shifting its peak. A change in industrial composition—for instance, the relative decline of manufacturing and rise of services—would shift the aggregate spectral peak, even absent any change in the shock process. This offers a structural explanation for the observation that business cycle properties have evolved over time in ways not easily attributed to changes in shock volatility alone (Beaudry and Portier, 2006).

6 Discussion

6.1 Limitations of the Gaussian specification

The Gaussian profit function (3) produces exactly symmetric cycles: booms and busts have equal duration. This is counterfactual—empirical business cycles feature slower expansions and sharper contractions. Under general hump-shaped profit functions (Proposition 3), cycles become asymmetric, with the degree of asymmetry determined by the skewness of $\pi(n)$ around n_p. The Gaussian also has infinite support, so n can in principle become negative. In practice, for $A \ll n_p$, this is quantitatively irrelevant.

6.2 Amplitude selection

In the deterministic model without entry costs, the amplitude A is a free parameter: the conservation law $E = -\frac{\varphi}{2}A^2$ admits a one-parameter family of orbits. Section 4.4 resolves this through entry-cost dissipation (Proposition 7). With entry cost $\kappa > 0$ and stochastic forcing $\sigma_\varepsilon > 0$, the balance of dissipation and forcing selects a unique stationary amplitude: $A_{\text{rms}} = \sigma_\varepsilon/\sqrt{2\eta\varphi\mu}$, depending on the ratio of shock volatility to entry costs—both observable. The frequency $\omega = \sqrt{\varphi/\mu}$ is invariant throughout this process (shifting negligibly to $\sqrt{\varphi/\mu - \eta^2/4}$ for small η), so amplitude selection operates without altering the cycle frequency.

6.3 Microfoundations of the timing friction

The timing friction is micro-founded in Section 2.3 from congestion in the entry matching process, yielding $\mu = (1-\alpha)/[\alpha(\delta n_p)^2]$ (Proposition 1). Cross-industry variation in μ arises from three structural sources: (i) variation in α (matching technology differs across industries); (ii) variation in δn_p (industries with higher turnover have smaller μ); (iii) variation in R (industries with scarcer entry resources have higher effective μ). The capital-per-employee proxy $\hat{\mu}$ used in Section 5.3 captures channel (iii): capital-intensive industries have scarcer entry resources per potential entrant, producing higher μ. Online Appendix C provides an alternative derivation from discrete-time proportional adjustment costs, showing the quadratic form arises generically.

6.4 Empirical evidence for hump-shaped profits

The hump-shaped profit landscape is the key maintained assumption. Empirical support comes from several sources. Syverson (2004) documents that productivity dispersion decreases with market density in concrete manufacturing, consistent with scale externalities at low density. Combes et al. (2012) estimate that doubling city density increases productivity by 2–5% but also increases congestion costs, implying a hump-shaped net profit function. Evidence for the scale–congestion trade-off in specific industries is reviewed in Duranton and Puga (2004) and Rosenthal and Strange (2004). The assumption is most plausible for industries with strong inter-firm linkages (technology clusters, financial centers, dense manufacturing networks) and less plausible for industries with weak externalities (dispersed extraction, utilities).

6.5 Relationship to Jovanovic (1982)

Jovanovic (1982) showed that temporal information about fixed idiosyncratic types generates the industry cross-section without aggregate shocks. The timing friction in the present model generates aggregate dynamics without aggregate shocks. The two mechanisms are complementary:

	Jovanovic (1982)	**Timing search**
Object	Fixed type θ_i	Time-varying $\pi(n(t))$
Friction	Signal noise	Timing friction μ
Cross-section	Rich (sizes, survival)	Minimal
Aggregates	Monotone convergence	$n_p + A\cos(\omega t)$

A synthesis—Jovanovic (1982) for endogenous exit (type discovery), timing search for entry timing (ironing along time)—would produce entry waves followed by shakeouts, with the entry–exit lag governed by learning speed.

6.6 Welfare cost of endogenous cycles

The conservation law provides a natural welfare measure. Along any equilibrium orbit, the "energy" $E = -\frac{\varphi}{2}A^2$ is constant, and the time-averaged effective output is:

$$\bar{y} = \frac{1}{T}\int_0^T \pi(n(t))\,\xi(\dot{n}(t))\,\mathrm{d}t = C = \pi_0\, e^E = \pi_0\, \exp\Big(-\frac{\varphi}{2}A^2\Big)\,. \tag{23}$$

Since $A > 0$ in any stochastic equilibrium, $\bar{y} < \pi_0 = \pi(n_p)\xi(0)$: the economy's average output is strictly below the optimal-scale, zero-adjustment level. The proportional welfare loss from cycling is:

$$\mathcal{L} = 1 - \frac{\bar{y}}{\pi_0} = 1 - e^{-\varphi A^2/2} \approx \frac{\varphi}{2} A^2 \quad \text{for small } A. \tag{24}$$

The welfare cost of business cycles in this model depends on two structural parameters: φ (the curvature of the profit landscape, which determines how costly it is to deviate from optimal scale) and A (the cycle amplitude, which in the stochastic equilibrium is determined by σ_ε and the shape of the potential well). Crucially, $\mathcal{L}$ does not depend on μ: the timing friction determines the *frequency* of cycles but not their welfare cost. Two industries with the same φ and A but different μ cycle at different frequencies yet suffer the same welfare loss—an industry that cycles slowly with large swings is equally costly as one that cycles quickly with the same amplitude.

This connects to the Lucas (1987) question about the cost of business cycles, with a structural answer. In the Lucas framework, the welfare cost depends on the variance of consumption, which is taken as a reduced-form object. Here, the welfare cost is a function of measurable primitives: $\mathcal{L} \approx \frac{\varphi}{2}A^2$, where φ can be estimated from the profit-landscape curvature and A from the observed amplitude of entry–exit cycles. The cost is zero only if $\sigma_\varepsilon = 0$ (no shocks to excite the center) or $\varphi = 0$ (flat profit landscape, no scale externalities). In the stochastic equilibrium, $\mathbb{E}[A^2]$ is proportional to $\sigma_\varepsilon^2/\varphi$, so the expected welfare loss scales as $\sigma_\varepsilon^2/2$—independent of the cycle frequency.

6.7 Open questions

Several extensions merit investigation. First, a multi-industry general equilibrium model in which industries with different (φ_i, μ_i) interact through factor markets would characterize the aggregate spectral density as an endogenous mixture. Second, the parameters φ and μ are taken as exogenous primitives; endogenizing them through industry investment in linkages (affecting φ) or entry infrastructure (affecting μ) would generate a richer set of predictions. Third, the interaction between the timing search mechanism and financial frictions (Beaudry et al., 2020) could produce models with multiple spectral peaks, one from market structure and one from financial amplification. Finally, the framework extends naturally to any setting where agents choose

when to act and face a trade-off between positive and negative externalities—hiring waves, investment cycles, technology adoption, and financial market dynamics.

7 Concluding Remarks

Business cycle frequency is a market-structure object. The timing search equilibrium implies $\omega = \sqrt{\varphi/\mu}$: cycles are faster in industries with sharper scale–congestion trade-offs and weaker timing frictions. This structural decomposition delivers cross-industry predictions, connects cycle frequency to independently measurable primitives, and provides a testable alternative to the view that spectral properties of fluctuations are determined by the shock process.

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